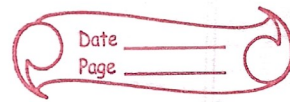


①

MATHSection-A

1) (a) $3 \times (0.05)' \times (0.95)^2$

2) $Q(x) = 300 - 50 + 10$
 $= 260$

3) (d) Neither

4) (a) Skew-symmetric matrix

5) (b) 4

6) $\cos^{-1}(\cos \frac{7\pi}{6})$

$$\begin{aligned} & \cos^{-1}[\cos(\pi + \frac{\pi}{6})] \quad \cos^{-1}[\cos(2\pi - \frac{5\pi}{6})] \\ & \cos^{-1}[\cos(-\frac{\pi}{6})] \quad \cos^{-1}[\cos(\frac{5\pi}{6})] \\ & \Rightarrow -\frac{\pi}{6} \quad \Rightarrow \frac{5\pi}{6} \end{aligned}$$

(b) $\frac{5\pi}{6}$

7) (b) (3, 4)

8) $P(A) = 0.4 \quad P(B/A) = 0.3 \quad P(B/A') = 0.5$

$P(A \cap B) = P(A) \cdot P(B/A) = 0.12$

$P(B/A') = \frac{P(B \cap A')}{P(A')} = 0.5 \Rightarrow P(B \cap A') = 0.5 \cdot P(A')$

$P(B \cap A') = 0.5 \cdot (1 - 0.4) = 0.3$

$P(B) = P(A \cap B) + P(B \cap A') = 0.12 + 0.3 = 0.42$

(c) 0.42

9) $f(x) = |x-3|$

(b) continuous but not differentiable at $x=3$

10) $\vec{a} \times \vec{b} = \vec{0} \quad \vec{a} \cdot \vec{b} = 0$

(a) $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$

11) $y = \log_e (\sin x)$

$$\frac{d}{dx} [\log_e (\sin x)] = \frac{1}{\sin x} \cdot \cos x$$

(b) $\cot x$

12) $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

$$A^3 = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

(b) $\begin{bmatrix} 8 & 0 \\ 0 & 27 \end{bmatrix}$

13) $y = \sin^{-1}(2x)$

$$-1 \leq 2x \leq 1$$

$$-\frac{1}{2} \leq x \leq \frac{1}{2}$$

(b) $\left[-\frac{1}{2}, \frac{1}{2}\right]$

14) (c) May or may not exist

15) $R = \{(5,5), (6,6)\}$
 $A = \{(4,6), (5,5), (5,6), (6,4), (6,5), (6,6)\}$

(a) $\frac{1}{3}$

16) $A = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$

$A^T = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$ $-A = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$

(a) Both are true & R is for A

17) (a) Both are true & R is for A

18) (c) A is true, R is false

19) $f(x) = e^x$ and $g(x) = \sqrt{x}$ at $x=1$

$\frac{dy}{dx} = \frac{f'(x)}{g'(x)} = \frac{e^x}{\frac{1}{2\sqrt{x}}} = 2\sqrt{x} e^x$

At $x=1$, $\Rightarrow 2e$

(a) Both are true & R is for A

20) (d) A is false & R is true

Q1

- 21) Let E_1 : Event that person has virus
 E_2 : Person doesn't have virus
 A : Test result is positive

(P) $P(E_1) = 0.01$ (1%)
 $P(E_2) = 1 - 0.01 = 0.99$ (99%)
 $P(A/E_1) = 0.95$ (95%)
 $P(A/E_2) = 0.02$ (2%)

(PP) $P(E_1/A) = \frac{P(E_1) \cdot P(A/E_1)}{P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2)}$
 $= \frac{0.01 \times 0.95}{0.01 \times 0.95 + 0.99 \times 0.02}$
 $= \frac{95}{95 + 198} = \frac{95}{293} \approx 0.324$
 $\therefore \boxed{32.42\%}$

22) (P) Maximize $Z = 400x + 500y$

Min

Subject to constraints:

- Machine time : $3x + 2y \leq 18$
- Finishing time : $1x + 2y \leq 10$
- Non-negativity : $x \geq 0, y \geq 0$

(P^o) For $3x + 2y = 18$

$x = 0, y = 9$ (0, 9)

$x = 6, y = 0$ (6, 0)

For $1x + 2y = 10$

$x = 0, y = 5$ (0, 5)

$x = 10, y = 0$ (10, 0)

$3x + 2y - x - 2y = 8$

$2x = 8 \Rightarrow \boxed{x = 4}$

5

Substitute x in $3x + 2y = 10$,

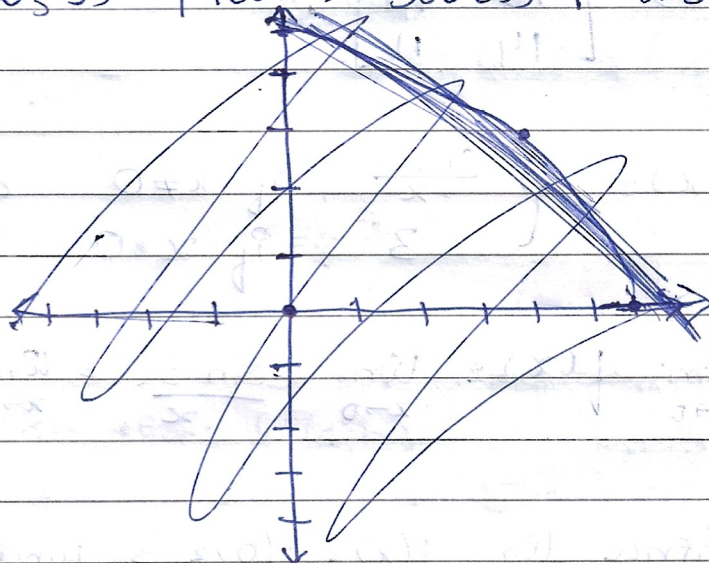
$$4 + 2y = 10$$

$$2y = 6$$

$$y = 3$$

Intersection point is $(4, 3)$

Corner points	Profit	Value (£)
$(0, 0)$	$400(0) + 500(0)$	0 (min)
$(6, 0)$	$400(6) + 500(0)$	2400
$(4, 3)$	$400(4) + 500(3)$	3100 (max)
$(0, 5)$	$400(0) + 500(5)$	2500



∴ Carpenter should produce 4 Bookshelves & 3 Tables to achieve max profit of £ 3,100.

23) $P(x) = -2x^2 + 120x - 500$

$$P'(x) = -4x + 120$$

Let $P'(x) = 0$

$$4x = 120$$

$$x = 30$$

$$P''(x) = -4 < 0 \text{ (confirms local max)}$$

∴ Manufacturer must sell 30 items.

24) $A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$

$A^2 - B = ?$

$\Rightarrow \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 4+12 & 8+8 \\ 6+6 & 12+4 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$

$\begin{bmatrix} 16 & 16 \\ 12 & 16 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 15 & 13 \\ 14 & 11 \end{bmatrix}$

25) $f(x) = \begin{cases} \frac{\sin 3x}{x}, & \text{if } x \neq 0 \\ 3, & \text{if } x = 0 \end{cases}$ at $x=0$

$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin 3x}{x} = \lim_{x \rightarrow 0} \frac{3 \sin 3x}{3x}$

$= 3 \times 1 = 3$

Since $\lim_{x \rightarrow 0} f(x) = f(0) = 3$, function is continuous at $x=0$.

26) $y = x^{\sin x}$

Taking natural log on both sides:

$\ln y = \sin x \cdot \ln x$

Differentiate

$\frac{1}{y} \frac{dy}{dx} = \cos x \ln x + \sin x \cdot \frac{1}{x}$

$\frac{dy}{dx} = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right)$

27) $\vec{a} = 3\hat{i} + 3\hat{j} - \hat{k}$ $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$

$$\vec{R} = \vec{a} + \vec{b}$$

$$\vec{R} = 3\hat{i} + \hat{j} + \hat{k}$$

$$|\vec{R}| = \sqrt{3^2 + 1^2 + 1^2} = \sqrt{11}$$

Unit vector: $\hat{u} = \frac{3\hat{i} + \hat{j} + \hat{k}}{\sqrt{11}}$

Vector of magnitude 5:

$$\vec{r} = \pm 5\hat{u} = \pm 5 \left(\frac{3\hat{i} + \hat{j} + \hat{k}}{\sqrt{11}} \right)$$

28) $R = \{(a, b) : a \leq b\}$

Reflexive: Since $a \leq a$ is true for all $a \in \mathbb{R}$,
 $\therefore R$ is reflexive

Transitive: If $(a, b) \in R$ & $(b, c) \in R$, then
 $a \leq b, b \leq c \Rightarrow a \leq c$, so $(a, c) \in R$
 $\therefore R$ is transitive

Symmetric: ~~Not~~ Take $a = 2, b = 3$

If $(a, b) \in R$, $2 \leq 3$, which is true

but $(b, a) \notin R$ as $3 \not\leq 2$, which is false

$\therefore R$ is not symmetric.

29) $\tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right)$

Divide by $\cos x$

$$\tan^{-1} \left(\frac{1 - \tan x}{1 + \tan x} \right) \Rightarrow \tan^{-1} \left(\frac{\tan(\pi/4) - \tan(x)}{\tan(\pi/4) + \tan(x)} \right)$$

$$\Rightarrow \tan^{-1} \left(\tan \left(\frac{\pi}{4} - x \right) \right)$$

$$\Rightarrow \boxed{\frac{\pi}{4} - x}$$

8

$$30) A = \begin{vmatrix} x+y & y+z & z+x \\ z & x & y \\ 1 & 1 & 1 \end{vmatrix} \quad R_1 \rightarrow R_1 + R_2$$

$$A = \begin{vmatrix} x+y+z & x+y+z & x+y+z \\ z & x & y \\ 1 & 1 & 1 \end{vmatrix}$$

$$A = (x+y+z) \begin{vmatrix} 1 & 1 & 1 \\ z & x & y \\ 1 & 1 & 1 \end{vmatrix}$$

Since R_1 & R_3 are equal, $\boxed{\Delta = 0}$ ✓

$$31) A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^3 - 3A^2 - A + 9I = 0$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix} A$$

$$= \begin{bmatrix} 1-2+1 & -1-1+1 & 1+3+1 \\ 2+2-3 & -2+1-3 & 2-3-3 \\ 1+2+1 & -1+1+1 & 1-3+1 \end{bmatrix} A$$

$$= \begin{bmatrix} 0 & -1 & 5 \\ 1 & -4 & -4 \\ 4 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0-2+5 & 0-1+5 & 0+3+5 \\ 1-8-4 & -1-4-4 & 1+12-4 \\ 4+2-1 & -4+1-1 & 4-3-1 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 8 \\ -11 & -9 & 9 \\ 5 & -4 & 0 \end{bmatrix}$$

$$-3A^2 = \begin{bmatrix} 0 & 3 & -15 \\ -3 & 12 & 12 \\ -12 & -3 & 3 \end{bmatrix}, -A = \begin{bmatrix} -1 & 1 & -1 \\ -2 & -1 & 3 \\ -1 & -1 & -1 \end{bmatrix}, 9I = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

9

Row 1: $[3+0-1+9] = 11 \neq 0$

$\therefore A^3 - 3A^2 - A + 9I = 0$

$A^{-1} :$

$A^3 - 3A^2 - A + 9I = 0$

$9I = -A^3 + 3A^2 + A$

Multiply by $A^{-1} :$

$9A^{-1} = -A^2 + 3A + I$

$A^{-1} = \frac{1}{9} (-A^2 + 3A + I)$

$= \frac{1}{9} \left(\begin{bmatrix} 0 & -1 & 5 \\ 1 & -4 & -4 \\ 4 & 1 & -1 \end{bmatrix} + \begin{bmatrix} 3 & -3 & 3 \\ 6 & 3 & -9 \\ 3 & 3 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)$

$= \frac{1}{9} \begin{bmatrix} 4 & -8 & 8 \\ 7 & 0 & -13 \\ 7 & 4 & 3 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 4 & -2 & -2 \\ 5 & 8 & -5 \\ -1 & 2 & 5 \end{bmatrix}$

32) $x = a(\theta - \sin \theta)$; $y = a(1 - \cos \theta)$

$\frac{dx}{d\theta} = a(1 - \cos \theta)$; $\frac{dy}{d\theta} = a \sin \theta$

$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{a \sin \theta}{a(1 - \cos \theta)}$

At $\theta = \frac{\pi}{2}$,

$\frac{dy}{dx} = \frac{\sin(\pi/2)}{1 - \cos(\pi/2)} = \frac{1}{1 - 0} = 1$

33) Bag 1: 4 Red, 5 Black (Total = 9)

Bag 2: 3 Red, 4 Black (Total = 7)

Let T_R : Transfer a red ball from Bag 1

T_B : Transfer a black ball from Bag 1

$$P(T_R) = \frac{4}{9}, \quad P(T_B) = \frac{5}{9}$$

Let R : Draw a red bag from Bag 2.

If red transferred: Bag 2 has 4R, 4B $\Rightarrow P(R/T_R) = \frac{4}{8}$

If black: Bag 2 has 3R, 5B $\Rightarrow P(R/T_B) = \frac{3}{8}$

$$\begin{aligned} P(R) &= P(R) = P(T_R) \cdot P(R/T_R) + P(T_B) \cdot P(R/T_B) \\ &= \frac{4}{9} \times \frac{4}{8} + \frac{5}{9} \times \frac{3}{8} = \frac{16 + 15}{72} = \frac{31}{72} \end{aligned}$$

$$ii) \quad P(T_B|R) = \frac{P(T_B) \cdot P(R/T_B)}{P(R)}$$

$$= \frac{\frac{5}{9} \times \frac{3}{8}}{\frac{31}{72}} = \frac{15}{31} //$$

34) $y = (\sin^{-1} x)^2 \quad (1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 2 = 0$

Differentiating w.r.t x : $y_1 = \frac{dy}{dx} = 2(\sin^{-1} x) \cdot \frac{1}{\sqrt{1-x^2}}$

$$\sqrt{1-x^2} y_1 = 2 \sin^{-1}(x)$$

Squaring $(1-x^2) y_1^2 = 4 (\sin^{-1}(x))^2 = 4y$

Differentiating w.r.t x :

$$(1-x^2) \cdot 2y_1 y_2 + y_1^2 (-2x) = 4y_1$$

Divide by $2y_1$:

$$(1-x^2) y_2 - x y_1 = 2$$

$$\Rightarrow (1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 2 = 0 //$$

35) $f(x) = \frac{4x+3}{5}$

One - one:

Let $f(x_1) = f(x_2)$
 $\frac{4x_1+3}{5} = \frac{4x_2+3}{5}$

$4x_1 = 4x_2$

$x_1 = x_2$

∴ One-one function.

Onto:

For any $y \in \mathbb{R}$, let $y = \frac{4x+3}{5}$

$5y = 4x + 3$

$x = \frac{5y-3}{4}$

Since $x \in \mathbb{R}$, ∴ Onto function.

∴ Bijective, thus, Invertible.

36)

Boundary lines:

$x+y=2$ (2,0) (0,2)

$x+3y=3$ (3,0) (0,1)

Corner points of feasible region:

(3,0) and (0,2)

~~$x+y=2$~~ ~~$x+3y=3$~~ $x+3y-x-y=1$
 $2y=1$

$y = \frac{1}{2}$

$x + \frac{1}{2} = 2 \Rightarrow x = 2 - \frac{1}{2} \Rightarrow x = \frac{3}{2}$

$\left(\frac{3}{2}, \frac{1}{2}\right)$

12

2:

$$\text{At } (3, 0), z = 3(3) + 5(0) \\ = 9$$

$$\text{At } (0, 2), z = 3(0) + 5(2) \\ = 10$$

$$\text{At } \left(\frac{3}{2}, \frac{1}{2}\right), z = 3\left(\frac{3}{2}\right) + 5\left(\frac{1}{2}\right) \\ = \frac{14}{2} = 7.$$

Minimum cost = 7.

37) Let centre of semicircle be the origin.

Let A & B be endpoints of diameter.

~~Let P be~~